

NETWORK DEPLOYMENT GUIDE

LISP IPv4 Standalone Deployment

IOS-XE 17.3.x • Three-Router Lab • IPv4 Overlay

Locator/ID Separation Protocol (LISP) overlay to isolate endpoint IP addresses (EIDs) from the physical transport network.

Platform: Cisco IOS-XE 17.3.x **Protocol:** LISP (RFC 6830)

Topology: 3 Routers (R1 xTR | R2 MS/MR | R3 xTR)

Scope: Layer 3 IPv4 EID-to-RLOC mapping, standalone

1. Overview & Objective

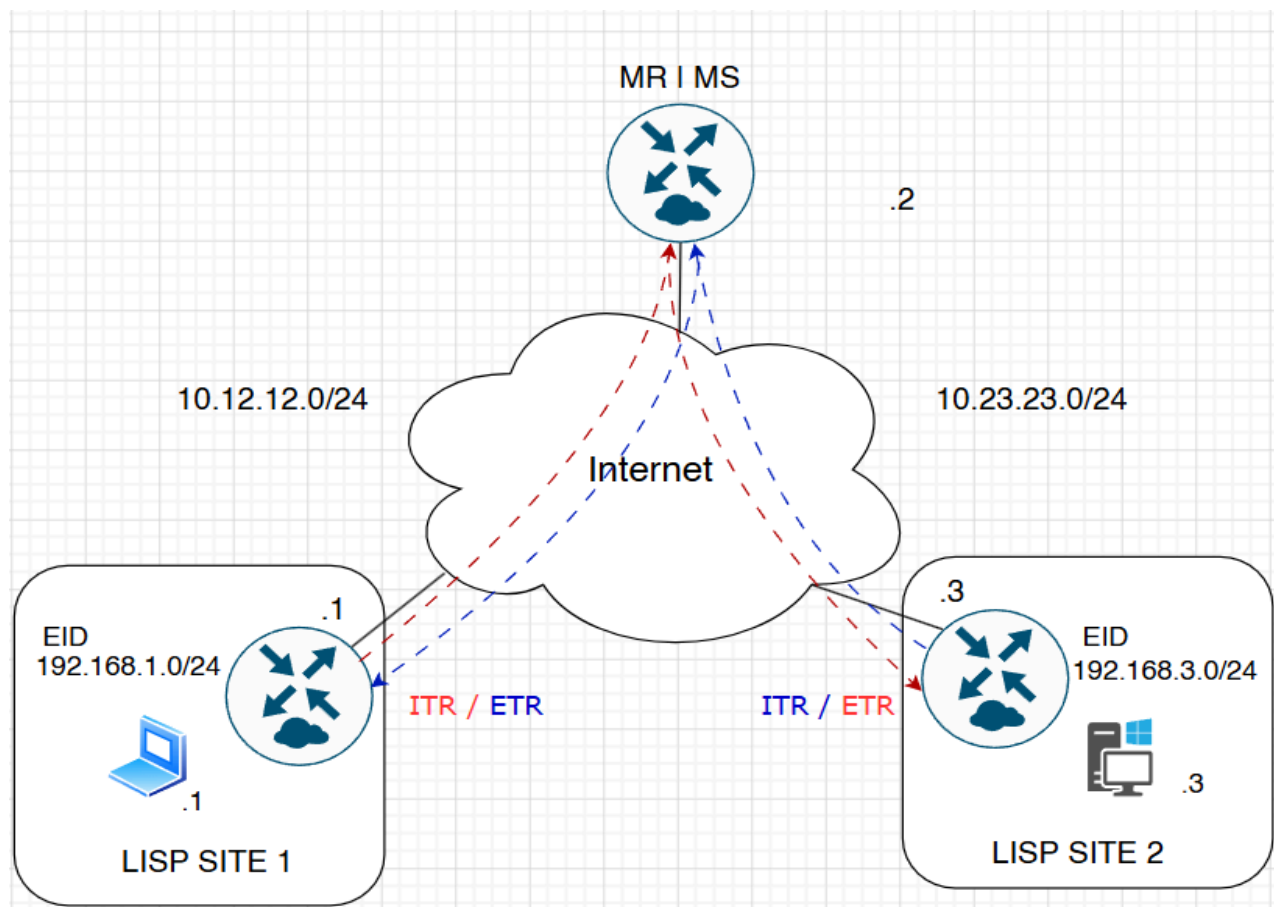
This guide walks through a complete Layer 3 LISP overlay deployment across a three-router lab topology running Cisco IOS-XE 17.3.x. The goal is to isolate endpoint IP addresses (EIDs) from the physical transport network (underlay) using the Locator/ID Separation Protocol.

What is LISP?

LISP separates the IP address space into two namespaces: Endpoint Identifiers (EIDs), which identify hosts, and Routing Locators (RLOCs), which identify routers in the underlay. This decoupling enables mobility, multi-homing, and scalable routing without exposing host subnets to the core.

2. Topology & Device Roles

The lab uses three routers connected in a linear chain. R2 sits in the core and provides both OSPF underlay routing and the LISP control plane (Map Server / Map Resolver). R1 and R3 act as edge xTRs for their respective sites.



Device	Role	Key IP Address(es)
R1	Site A — Ingress/Egress Tunnel Router (xTR)	Loopback0: 192.168.1.1 / Gi1: 10.12.12.1
R2	Core — Map Server / Map Resolver (MS/MR)	Gi1: 10.12.12.2 / Gi2: 10.23.23.2
R3	Site B — Ingress/Egress Tunnel Router (xTR)	Loopback0: 192.168.3.3 / Gi1: 10.23.23.3

Key address spaces:

Prefix	Type	Description
10.12.12.0/24	RLOC (Underlay)	R1 — R2 transit link
10.23.23.0/24	RLOC (Underlay)	R2 — R3 transit link
192.168.1.0/24	EID (Overlay)	Site A endpoint space (R1)
192.168.3.0/24	EID (Overlay)	Site B endpoint space (R3)

3. Phase 1 — R1 Configuration (Site A xTR)

R1 is the edge router for Site A. It hosts the EID prefix 192.168.1.0/24 on its Loopback0 interface, registers this prefix with the Map Server (R2), and encapsulates/decapsulates LISP traffic for the site.

R1 Roles

ITR (Ingress Tunnel Router): encapsulates outbound traffic toward EID destinations after resolving the RLOC via map-request. ETR (Egress Tunnel Router): decapsulates inbound LISP-encapsulated traffic and delivers to the local EID.

3.1 Interface Configuration

Loopback0 represents the Site A endpoint EID. GigabitEthernet1 is the uplink RLOC interface toward R2.

```
interface Loopback0
  description SITE_A_ENDPOINT_EID
  ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet1
  ip address 10.12.12.1 255.255.255.0
  negotiation auto
  no mop enabled
  no mop sysid
  no shutdown
```

3.2 Underlay Routing (OSPF)

Only the RLOC network is advertised into OSPF. The EID (192.168.1.0/24) is intentionally kept out of OSPF — it remains invisible to the core.

```
router ospf 1
  network 10.12.12.0 0.0.0.255 area 0
```

3.3 LISP Configuration

The LISP configuration defines a locator-set with R1's RLOC address, maps the EID prefix to that locator, and points to R2 as both the Map Resolver (for lookups) and the Map Server (for registration).

```
router lisp
  locator-set RLOC_R1
    10.12.12.1 priority 1 weight 100
  exit-locator-set
  !
  instance-id 0
    service ipv4
      eid-table default
      database-mapping 192.168.1.0/24 locator-set RLOC_R1
      itr map-resolver 10.12.12.2
    itr
      etr map-server 10.12.12.2 key cisco123
    etr
  exit-service-ipv4
exit-instance-id
exit-router-lisp
```

Authentication Key

The key cisco123 must match exactly on both the xTR (etr map-server ... key) and the MS/MR (authentication-key). A mismatch will cause registration failures.

4. Phase 2 — R2 Configuration (Core MS/MR)

R2 is the network core. It provides OSPF underlay routing between the two sites and operates as the LISP control plane: accepting Map-Register messages from xTRs and answering Map-Request queries. It holds the EID-to-RLOC mapping database.

R2 Roles

Map Server (MS): accepts Map-Register messages from ETRs and stores EID-to-RLOC mappings. Map Resolver (MR): receives Map-Request messages from ITRs and returns Map-Reply with the RLOC.

4.1 Interface Configuration

R2 has two uplinks — one toward R1 and one toward R3 — to bridge both transit segments.

```
interface GigabitEthernet1
 ip address 10.12.12.2 255.255.255.0
 negotiation auto
 no mop enabled
 no mop sysid
 no shutdown
!
interface GigabitEthernet2
 ip address 10.23.23.2 255.255.255.0
 negotiation auto
 no mop enabled
 no mop sysid
 no shutdown
```

4.2 Underlay Routing (OSPF)

Both transit subnets are advertised into OSPF. The EID prefixes (192.168.x.x) are never injected here — this is the fundamental LISP design principle.

```
router ospf 1
 network 10.12.12.0 0.0.0.255 area 0
 network 10.23.23.0 0.0.0.255 area 0
```

4.3 LISP Control Plane

R2 defines authorized EID prefixes for each site and binds them with the shared authentication key. The `ipv4 map-server` and `ipv4 map-resolver` commands activate both control-plane functions.

```
router lisp
  site SITE_A
    authentication-key cisco123
    eid-prefix 192.168.1.0/24
  exit
!
  site SITE_B
    authentication-key cisco123
    eid-prefix 192.168.3.0/24
  exit
!
  ipv4 map-server
  ipv4 map-resolver
exit
```

5. Phase 3 — R3 Configuration (Site B xTR)

R3 mirrors R1's role for Site B. It hosts the EID prefix 192.168.3.0/24 on Loopback0, registers with R2, and handles LISP encapsulation for outbound traffic toward Site A.

5.1 Interface Configuration

```
interface Loopback0
  description SITE_B_ENDPOINT_EID
  ip address 192.168.3.3 255.255.255.0
!
interface GigabitEthernet1
  ip address 10.23.23.3 255.255.255.0
  negotiation auto
  no mop enabled
  no mop sysid
  no shutdown
```

5.2 Underlay Routing (OSPF)

```
router ospf 1
  network 10.23.23.0 0.0.0.255 area 0
```

5.3 LISP Configuration

R3's LISP config is symmetric to R1's, with its own RLOC address (10.23.23.3) and pointing to R2 via the 10.23.23.2 address for map-resolver and map-server.

```
router lisp
  locator-set RLOC_R3
    10.23.23.3 priority 1 weight 100
  exit-locator-set
!
instance-id 0
  service ipv4
    eid-table default
```



```
database-mapping 192.168.3.0/24 locator-set RLOC_R3
itr map-resolver 10.23.23.2
itr
etr map-server 10.23.23.2 key cisco123
etr
exit-service-ipv4
exit-instance-id
exit-router-lisp
```

6. Operational Verification

Execute these checks in order after completing all three phases. Each check builds on the previous one: underlay first, then control plane, then data plane.

Step	Command	Router	Expected Result
1	<code>show ip route ospf</code>	R2	10.12.12.0/24 and 10.23.23.0/24 present — 192.168.x.x subnets absent (EIDs hidden from core)
2	<code>show lisp site</code>	R2	Both SITE_A and SITE_B display 'Up: yes' — xTRs registered successfully
3	<code>ping 192.168.3.3 source loopback 0</code>	R1	Success — LISP encapsulation and forwarding working end-to-end
4	<code>show ip lisp map-cache</code>	R1	Entry for 192.168.3.0/24 with RLOC 10.23.23.3 — dynamically resolved and cached

6.1 Verification Deep Dive

Step 1 — Underlay Routing

The absence of 192.168.x.x routes in the OSPF table on R2 confirms the EID-RLOC separation is working correctly. If EID prefixes appear in OSPF, check that no redistribution is configured under router ospf 1 on either xTR.

Step 2 — Control Plane Registration

The show lisp site output on R2 shows each registered site, its EID prefix, the registering RLOC, and registration uptime. An 'Up: no' result typically indicates an authentication-key mismatch or an OSPF reachability issue between the xTR's RLOC and R2.

Step 3 — Data Plane End-to-End

The first ping from R1 to 192.168.3.3 triggers a Map-Request toward R2. R2 responds with a Map-Reply containing R3's RLOC (10.23.23.3). R1 then LISP-encapsulates the packet (outer header: 10.12.12.1 → 10.23.23.3) and forwards it through the underlay.

Step 4 — Map Cache Validation

After a successful ping, the map-cache on R1 should contain a dynamic entry for 192.168.3.0/24 pointing to RLOC 10.23.23.3 with priority 1, weight 100. Cache entries have a default TTL of 24 hours and are refreshed by subsequent traffic.

7. Troubleshooting Reference

Symptom	Likely Cause	Fix
show lisp site shows Up: no	Authentication-key mismatch	Verify cisco123 matches on xTR etr map-server and R2 authentication-key
Map cache empty after ping	OSPF reachability failure	Confirm RLOC subnets (10.12.12.x, 10.23.23.x) are in OSPF table on R2
Ping fails despite registration	EID not in database-mapping	Check database-mapping prefix matches the site's eid-prefix on R2
LISP registration loops	Duplicate instance-id	Confirm instance-id 0 used consistently across all routers

Debug Commands

debug lisp control-plane — traces Map-Register and Map-Request/Reply exchanges
debug lisp data-plane — traces LISP encapsulation decisions
show lisp session — shows active LISP control-plane sessions and their state

R1

```
config terminal
hostname R1
!
interface Loopback0
description SITE_A_ENDPOINT_EID
ip address 192.168.1.1 255.255.255.0
!
interface LISP0
!
interface GigabitEthernet1
ip address 10.12.12.1 255.255.255.0
negotiation auto
no mop enabled
no mop sysid
!
router lisp
locator-set RLOC_R1
10.12.12.1 priority 1 weight 100
exit-locator-set
!
instance-id 0
service ipv4
eid-table default
database-mapping 192.168.1.0/24 locator-set RLOC_R1
itr map-resolver 10.12.12.2
itr
etr map-server 10.12.12.2 key cisco123
etr
exit-service-ipv4
!
exit-instance-id
!
exit-router-lisp
!
router ospf 1
network 10.12.12.0 0.0.0.255 area 0
```

R2

Config terminal

Hostname R2

!

interface GigabitEthernet1

ip address 10.12.12.2 255.255.255.0

negotiation auto

no mop enabled

no mop sysid

!

interface GigabitEthernet2

ip address 10.23.23.2 255.255.255.0

negotiation auto

no mop enabled

no mop sysid

!!

router lisp

site SITE_A

authentication-key cisco123

eid-prefix 192.168.1.0/24

exit

!

site SITE_B

authentication-key cisco123

eid-prefix 192.168.3.0/24

exit

!

ipv4 map-server

ipv4 map-resolver

exit

!

router ospf 1

network 10.12.12.0 0.0.0.255 area 0

network 10.23.23.0 0.0.0.255 area 0

R3

Config terminal

Hostname R3

```
!  
!  
interface Loopback0  
description SITE_B_ENDPOINT_EID  
ip address 192.168.3.3 255.255.255.0  
!  
interface LISP0  
!  
interface GigabitEthernet1  
ip address 10.23.23.3 255.255.255.0  
negotiation auto  
no mop enabled  
no mop sysid  
!!  
router lisp  
locator-set RLOC_R3  
10.23.23.3 priority 1 weight 100  
exit-locator-set  
!  
instance-id 0  
service ipv4  
eid-table default  
database-mapping 192.168.3.0/24 locator-set RLOC_R3  
itr map-resolver 10.23.23.2  
itr  
etr map-server 10.23.23.2 key cisco123  
etr  
exit-service-ipv4  
!  
exit-instance-id  
!  
exit-router-lisp  
!  
router ospf 1  
network 10.23.23.0 0.0.0.255 area 0
```

